

Econometrics - Project

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Project Title	Educational Attainment and Occupational Skill Upgrading in the Netherlands: An OLS Study Using ISCO Codes
Abstract (100 words max)	This project investigates the relationship between educational attainment and occupational skill level in the Netherlands using IPUMS International microdata from 2011. Individuals are assigned to high-skill or low-skill occupations based on ISCO classification, allowing the construction of a skill indicator. OLS regression is used to estimate how education affects the probability of holding a high-skill job, controlling for age, gender, marital status, and nativity. The analysis contributes to understanding human-capital allocation and skill upgrading within the Dutch labor market.

Sample selection and descriptive statistics – 300 words max

Word count: 282

This project uses microdata from the IPUMS International database, focusing on individuals residing in the Netherlands in 2011. The dataset is a cross-sectional census extract, and the unit of analysis is the individual, uniquely identified by serial and pernum. The sampling frame covers the working-age population, restricted to individuals aged 17 to 67, reflecting the typical entry age into the labor force and the statutory retirement age in the Netherlands.

The final analysis sample includes 231,986 individuals. Cleaning procedures were deliberately minimal. Observations with missing values in any of the core variables—age, sex, education, or occupation—were removed to ensure consistency across the analytical sample. Age (age) is measured in completed years. Sex (sex) is coded as 1 for males and 2 for females. Educational attainment (educn1) follows the IPUMS international classification, where higher values represent higher levels of schooling. The variable HighSkill is a binary indicator derived from ISCO occupation codes (occisco), equal to 1 for high-skill occupations and 0 otherwise. No trimming or winsorization was applied, as the selected variables do not exhibit extreme or implausible values. All descriptive statistics are computed using the unweighted sample.

Variable	Obs	Mean	Std. dev.	Min	Max
age	231,986	40.32494	13.30097	17	67
sex	231,986	1.471593	.4991935	1	2
educn1	231,986	289.251	98.341	0	998
HighSkill	231,986	.434884	.4957428	0	1

Exhibit 1 reports summary statistics for age, sex, education, and high-skill status. The average age is around 40 years, and the standard deviation indicates substantial variation within the working-age population. Roughly 43

percent of individuals are employed in high-skill occupations, and education levels display considerable heterogeneity.

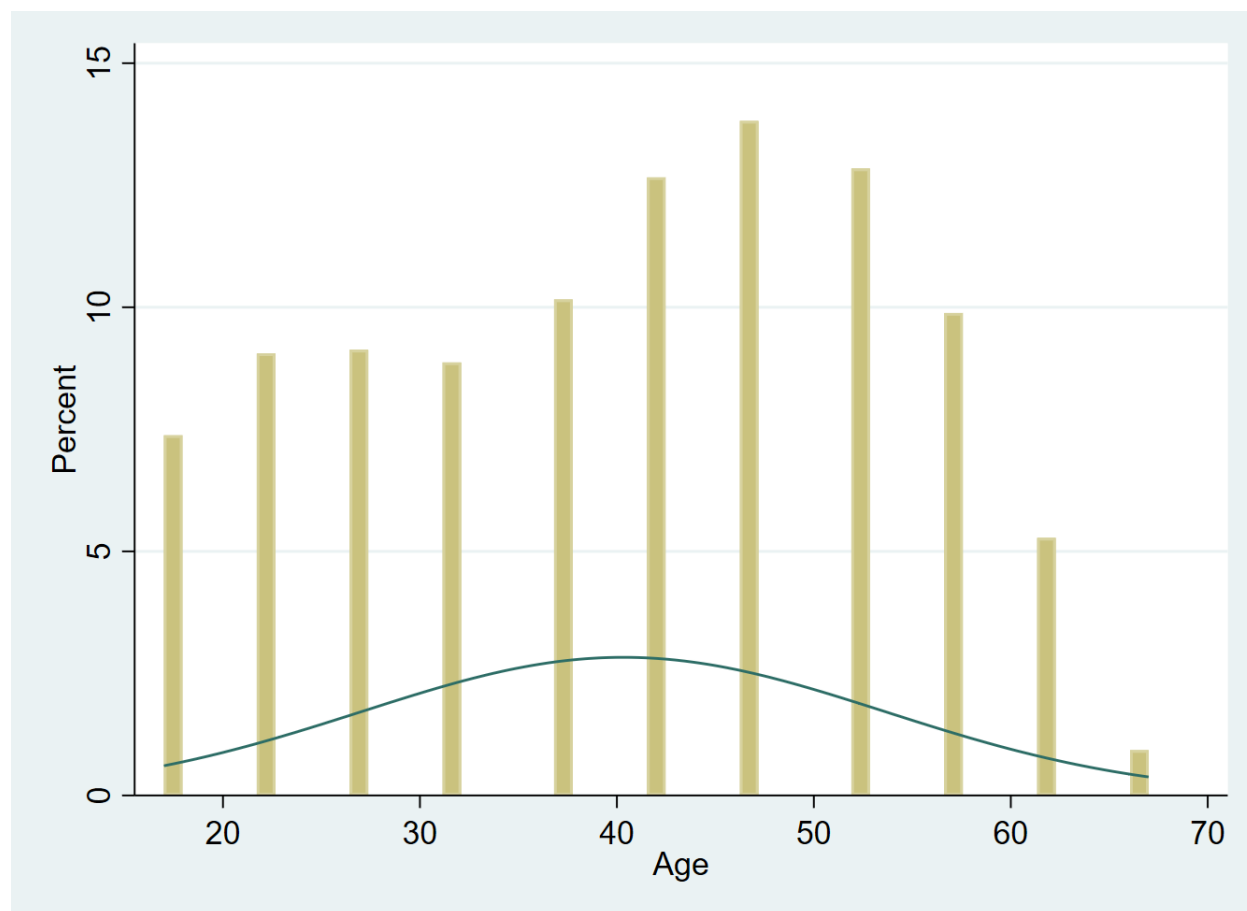


Exhibit 2 presents the distribution of age. The histogram is concentrated between ages 25 and 55 and includes a normal density curve for comparison. This figure complements the summary statistics by illustrating the demographic structure of the final analysis sample.

Hypothesis testing and regression model – 600 words max

Word count: 577

This section presents the hypotheses, econometric model, and identification strategy used to analyse how educational attainment predicts the probability of holding a high-skill occupation in the Netherlands. The dependent variable is a binary indicator equal to one for individuals employed in ISCO major groups 1–3 (managers, professionals, and technicians/associate professionals) and zero otherwise. The purpose of this analysis is to quantify how education shapes occupational sorting within the Dutch labour market.

Hypotheses

H1: Higher educational attainment increases the probability of holding a high-skill occupation ($\beta > 0$).

H2: Foreign-born individuals have a lower probability of holding a high-skill occupation, conditional on education and demographic characteristics ($\gamma < 0$).

These hypotheses are motivated by human capital theory, which predicts that education raises productivity and enables access to more complex occupations.

Outcome variable, key regressors, and controls

The outcome variable $HighSkill_i$ is a binary indicator that takes value 1 for high-skill occupations. The key explanatory variable is educational attainment (EDUCNL), coded as categorical indicators relative to “no education.” Control variables include age (continuous), sex (female indicator), marital status (categorical), and nativity (foreign-born indicator). These variables capture observable characteristics correlated with both education and occupational outcomes.

Econometric models

The main econometric specification is a Linear Probability Model (LPM), estimated by ordinary least squares:

$$HighSkill_i = \sum_k \beta_k 1\{Education_i = k\} + \gamma Age_i + \delta' Z_i + \varepsilon_i,$$

where Z_i denotes the vector of controls (sex, marital status, nativity). Coefficients in the LPM are interpreted as percentage-point changes in the probability of holding a high-skill occupation relative to the baseline education category.

Because the dependent variable is binary, a logistic regression model is estimated as a robustness check:

$$Pr(HighSkill_i = 1 | X_i) = \Lambda(\beta X_i),$$

where $\Lambda(\cdot)$ is the logistic cumulative distribution function. After estimation, predictive marginal effects are computed to express results in terms of probability differences across education categories, allowing direct comparison with OLS coefficients.

Identification strategy

Identification relies on cross-sectional variation in educational attainment across individuals aged 17–67. Since education is typically completed before individuals enter the labour market, reverse causality from current occupation to educational attainment is unlikely. The inclusion of demographic controls mitigates omitted-variable bias arising from observable characteristics that jointly influence education and occupational outcomes.

However, several threats to identification remain. Unobserved factors—such as ability, motivation, socioeconomic background, and access to labour-market networks—may influence both schooling choices and occupational sorting. If such unobservables are positively correlated with educational attainment and the likelihood of entering a high-skill occupation, the estimated education coefficients may be biased upward. Because the dataset is cross-sectional and does not include instruments or panel variation, these unobserved differences cannot be fully eliminated. For this reason, the estimated coefficients are interpreted as conditional associations rather than causal effects.

The identification strategy therefore rests on the assumption that, after conditioning on age, sex, marital status, and nativity, the remaining variation in educational attainment captures differences in human capital relevant for occupational choice. While this assumption is strong, the comparison between the LPM and logit models allows an assessment of robustness to functional-form assumptions. Consistency between the two models strengthens confidence that the estimated patterns reflect systematic relationships rather than artefacts of a specific specification.

Justification for robustness model

The logistic model is used to address the nonlinearity of the binary dependent variable and to verify that the estimated probability differences across education levels are not sensitive to the linearity assumption embedded in the LPM. The use of marginal effects ensures comparability between the two approaches. The robustness exercise is therefore motivated by functional-form considerations rather than by an alternative identification strategy.